

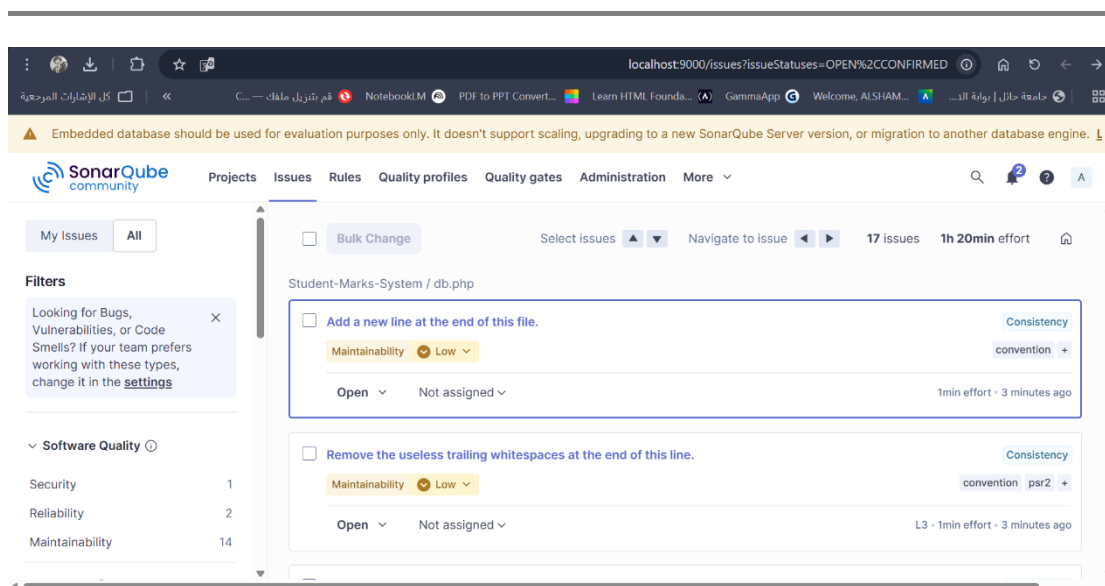
Software Quality Assurance Report

1. Introduction

In this project, I used SonarQube to analyze my "Student Marks System" project. The goal was to find and fix technical issues in the PHP and SQL code to improve the overall quality.

2. Project Details

- **Project Name:** Student-Marks-System.
- **Name:** Faisal Abdulkakem Alshammari.
- **Student ID:** 202205505.
- **Languages:** PHP, SQL, CSS.



3. Initial Scan Results

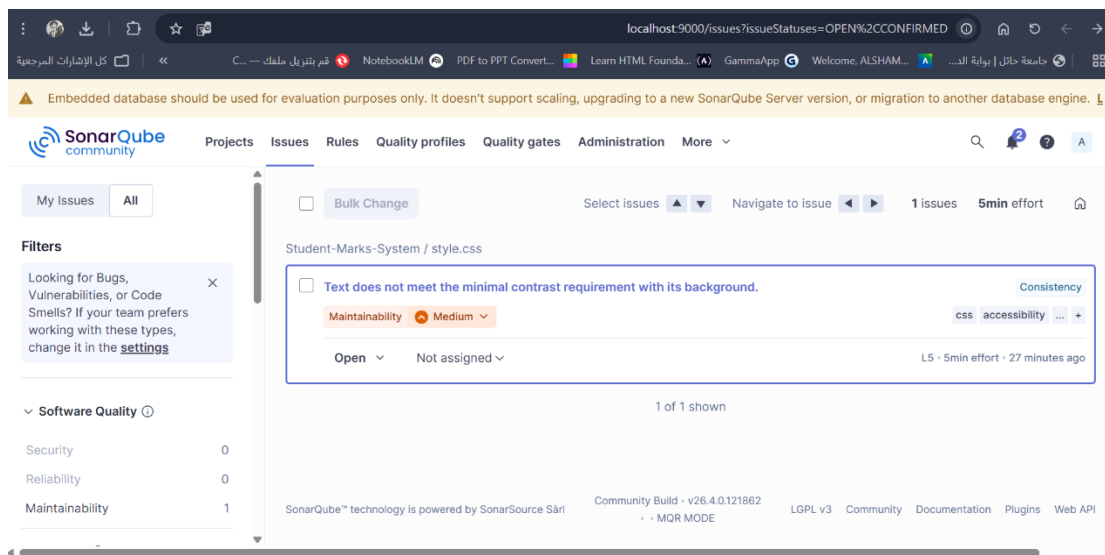
The first scan showed the following issues:

- **Security:** 1 Blocker (Database connection issue).
- **Reliability:** 2 Bugs.
- **Maintainability:** 14 Code Smells.

4. Applied Fixes (10 Changes)

I modified the code files (db.php, delete.php, and index.php) to fix these issues:

1. Added a password to the database connection for security.
2. Changed include to include_once in delete.php.
3. Changed include to include_once in index.php.
4. Added curly braces { } to if statements in index.php.
5. Removed the closing ?> tag in db.php to prevent errors.
6. Removed the closing ?> tag in delete.php.
7. Added a new empty line at the end of db.php.
8. Added a new empty line at the end of delete.php.
9. Removed extra spaces (trailing whitespaces) in db.php.
10. Removed extra spaces in index.php.



5. Final Results

After fixing the code and running the scanner again, the results improved significantly:

- **Security Issues: 0.**
- **Reliability Issues: 0.**
- **Maintainability Issues: 1** (Minor CSS issue).

6. Conclusion

This task helped me understand how to use professional tools like SonarQube to find hidden bugs in my code. Fixing these 10 issues made my "Student Marks System" more secure and better organized.